INCH-POUND

MIL-F-50541A (AR)
AMENDMENT 6
01 October 1996
SUPERSEDING
AMENDMENT 5
27 June 1994

MILITARY SPECIFICATION

FUZE FOR FLECHETTE WARHEAD, WDUA/A (LESS EXPULSION CHARGE)

This amendment forms a part of MIL-F-50541A (AR), dated 8 September 1978, and is approved for use by the US Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1: Delete "MIL-P-38633 - Primer, Stab, for WDU-4A/A Warhead." and substitute "MIL-P-63297 - Primer, Stab, PA515."

PAGE 2

2.1: Delete "65C11051" and substitute "9298753".

PAGE 3

3.2: Delete "65C11051" and substitute "9298753".

3.4.2: Delete "1.15 seconds" and substitute "1.20 seconds".

Delete "31.0 g's" and substitute "35.00 g's".

PAGE 4

3.6.1: Delete "High Temperature Procedure I Method 501 of MIL-STD-810" and substitute "4.5.10".

3.6.2: Delete "Low Temperature Procedure I at 54°C (-65°F) Method 502 of MIL-STD-810" and substitute "4.5.11".

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PAGE 4 (continued)

3.6.3: Delete "shock pulse with a peak of 44 plus or minus 4 g's and time duration of .13 plus or minus .002 seconds." and substitute "shock pulse with

(a) A peak of 44 plus or minus 4 g's and time duration of .013 plus or minus .002 seconds or

(b) A peak of 48 plus or minus 8 g's and time duration of .012 plus or minus .002 seconds."

PAGE 7

4.3.1: Delete "65B11049" and substitute "9298751".

Delete "65C11048" and substitute "9298750".

Delete "65C11047" and substitute "9298742".

Delete "65B11037" and substitute "9298747".

Delete "65C11052" and substitute "9298754".

Delete "65C11053" and substitute "9298755".

Delete "65C11054" and substitute "9298756".

Delete "65B11055" and substitute "9298757".

Delete "65C11057" and substitute "9298759".

Delete "65C11058" and substitute "9298760".

Delete "65C11051" and substitute "9298753".

Delete "65C11056" and substitute "9298758".

Delete "66B10082" and substitute "8883713".

Delete "66B10066" and substitute "8883718".

Delete "8883719" and substitute "9299424".

Delete "9239407" and substitute "9327030".

Delete "8883706" and substitute "9260776".

Delete "8883693" and substitute "9260804".

Delete "8883692" and substitute "9260774".

Delete "9239418" and substitute "9327657".

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PAGE 9

Table I: Delete "65B11049" and substitute "9298751".

Delete "65C11048" and substitute "9297850".

Delete "65C11047" and substitute "9298742".

Delete "65B11037" and substitute "9298747".

Delete "65C11052" and substitute "9298754".

Delete "65C11051" and substitute "9298753".

Delete "4.3.2.1" and substitute "4.4.2.1".

Delete "4.3.2.4" and substitute "4.4.2.4".

Delete "4.3.2.5" and substitute "4.4.2.5".

Delete "4.3.2.6" and substitute "4.4.2.6".

Delete "4.3.2.7" and substitute "4.4.2.7".

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Table I: Delete "65C11053" and substitute "9298755".

Delete "65C11054" and substitute "9298756".

Delete "65C11056" and substitute "9298758".

Delete "66B10082" and substitute "8883713".

Delete "66B10066" and substitute "8883718".

Delete "65C11051" and substitute "9298753".

PAGE 11

Table I: Delete "65C11055" and substitute "9298757".

Delete "65C11057" and substitute "9298759".

Delete "65C11058" and substitute "9298760".

Delete "65C11051" and substitute "9298753" in 2 places.

Delete "4.3.2.10" and substitute "4.4.2.10".

Delete "4.3.2.11" and substitute "4.4.2.11".

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PAGE 11 (continued)

Delete "4.3.2.12" and substitute "4.4.2.12".

Delete "4.3.2.19" and substitute "4.4.2.15".

Delete "Class" and substitute "Less".

Delete "(Dwg. 65811051)" and substitute "(Dwg. 9298753)".

PAGE 12

Table I: Delete "65C11051" and substitute "9298753".

PAGE 13

Table I: Delete "65C11051" and substitute "9298753".

Delete "8883706" and substitute "9260776".

Delete "8883698" and substitute "9260804".

PAGE 14

Table I: Delete "65C11051" and substitute "9298753".

Delete "8883719" and substitute "9298424".

Delete "9239418" and substitute "9327657".

Delete "8883692" and substitute "9260774".

PAGE 15

Table II: Delete "65C11048" and substitute "9298750".

Delete "65C11051" and substitute "9298753".

PAGE 16

Table II: Delete "65C11047" and substitute "9298742".

Delete "65C11051" and substitute "9298753".

PAGE 18

4.4.2.1: Delete "65C11049" and substitute "9298751".

PAGE 19

4.4.2.2: Delete "65C11048" and substitute "9298750".

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PAGE 20

4.4.2.3: Delete "65C11047" and substitute "9298742".

PAGE 21

4.4.2.4: Delete "65B11037" and substitute "9298747".

PAGE 22

4.4.2.5: Delete "9239407" and substitute "9327030".

PAGE 24

4.4.2.7: Delete "65C11052" and substitute "9298754".

Delete "65C11051" and substitute "9298753".

PAGE 25

4.4.2.8: Delete "65C11053" and substitute "9298755".

Delete "65C11051" and substitute "9298753".

Delete "Perpendicularity of locking ball holes to small outside diameter" and substitute "True position of locking ball holes to small inside diameter".

Add Major 109 as follows:

"109 True position of diameter of S&A cavity to small inside diameter 0.40% 3.2 Gage".

PAGE 26

4.4.2.9: Delete "65C11054" and substitute "9298756".

Delete "65C11051" and substitute "9298753".

Delete Major 103 in its entirety.

PAGE 27

4.4.2.10: Delete "65B11055" and substitute "9298757".

Delete "65C11051" and substitute "9298753".

PAGE 28

4.4.2.10: Delete "65C11057" and substitute "9298759".

Delete "65C11051" and substitute "9298753".

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PAGE 29

4.4.2.10: Delete "65B11058" and substitute "9298760".

Delete "65C11051" and substitute "9298753".

PAGE 30

4.4.2.13: Delete "65C11051" and substitute "9298753".

PAGE 31

4.4.2.14: Delete "9239418" and substitute "9327657".

PAGE 32

4.4.2.15: Delete "65C11051" and substitute "9298753".

PAGE 33

4.4.2.16: Delete "65C11056" and substitute "9298758".

Delete "65C11051" and substitute "9298753".

PAGE 34

4.4.2.17: Delete "66B10082" and substitute "8883713".

Delete "65C11051" and substitute "9298753".

PAGE 35

4.4.2.18: Delete "65B11067" and substitute "8883718".

Delete "65C11051" and substitute "9239423".

PAGE 36

4.4.2.19: Delete "8883692" and substitute "9260774".

PAGE 37

4.4.2.20: Delete "8883692" and substitute "9260774".

Under Category and Before Maximum gap between gear and pinion: Add "2".

PAGE 38

4.4.2.21: Delete "8883692" and substitute "9260774".

Under Category and Before Maximum gap between gear and pinion: Add "2".

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PAGE 39

4.4.2.22: Delete "8883706" and substitute "9260776".

Delete "8883692" and substitute "9260774".

PAGE 40

4.4.2.23: Delete "65C11046" and substitute "9239423".

Delete "4.5.7.6" and substitute "4.5.4.6".

Delete "4.5.7.7" and substitute "4.5.4.7".

PAGE 41

4.4.2.24: Delete "8883693" and substitute "9260804".

Delete "65C11046" and substitute "9260774".

PAGE 42

4.4.2.25: Delete "8883719" and substitute "9239424".

Delete "65C11046" and substitute "9239423".

PAGE 43

4.4.2.26: Delete "8883692" and substitute "9260774".

Delete "65C11056" and substitute "9239423".

PAGE 47

4.5.3.1: Delete "4.3.4" and substitute "4.4.4".

Last Line: Delete "the".

PAGE 48

4.5.4.5, Line 5: Delete "the required force" and substitute "the required force, the height of the lever pallet above the lever shall be inspected".

PAGE 50

4.5.5.1: Delete "4.3.4" and substitute "4.4.4".
4.5.5.2: Delete "4.3.4" and substitute "4.4.4".
4.5.5.3: Delete "4.3.4" and substitute "4.4.4".

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PAGE 51

4.5.5.4: Delete "4.3.4" and substitute "4.4.4".

4.5.5.5: Delete "4.3.4" and substitute "4.4.4".

Line 4: Delete "rte" and substitute "rate".

Delete "31.00 g's" and substitute "35.00 g's".

4.5.6.1: Delete "4.3.4" and substitute "4.4.4".

4.5.6.2: Delete "4.3.4" and substitute "4.4.4".

PAGE 52

4.5.7, Line 7: Delete "than" and substitute "then".

PAGE 53

4.5.10: Delete in its entirety and substitute:

"4.5.10 High temperature. This test is intended to approximate the exposure and determine the resistance of the fuze assembly to a high temperature storage condition for a period of time prior to operation. The fuze assembly will be placed in an environmental chamber with the internal chamber temperature raised to 71°C (160°F). The time of exposure to this temperature shall be a minimum of four (4) hours. The fuze shall be armed and fired within five minutes of removal from extreme temperature."

4.5.11: Delete in its entirety and substitute:

"4.5.11 Low temperature. This test is intended to approximate the exposure and determine the resistance of the fuze assembly to a low temperature storage condition for a period of time prior to operation. The fuze assembly will be placed in an environmental chamber with the internal chamber temperature lowered to -54°C (-65°F). The time of exposure to this temperature shall be a minimum of four (4) hours. The fuze shall be armed and fired within five minutes of removal from extreme temperature."

4.5.11: Add

"Note: Removing the fuze from the cold temperature environment prior to performing the centrifuge test may cause frost to form on the metal parts. This may have an effect on the test results due to attenuation of the acoustic signal of the arming mechanism, thus making it difficult to distinguish the arm signal from the background noise. The contractor shall take the appropriate action to alleviate this potential problem.

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One effective method has been to seal each fuze in a plastic bag prior to temperature conditioning. Bagged fuzes are then subjected to the centrifuge test. Prior to assembly into bags, maintain fuze assemblies in a 50% max. relative humidity under ambient temperature conditions of 70 ± 10 degrees F for a period of four hours. Hermetically (Heat) seal bags."

4.5.12: Delete "shock pulse with a peak of 44 plus or minus 4 g's and time duration of .13 plus or minus .002 seconds." and substitute "shock pulse with

(a) A peak of 44 plus or minus 4 g's and time duration of .013 plus or minus .002 seconds or

(b) A peak of 48 plus or minus 8 g's and time duration of .012 plus or minus .002 seconds."

PAGE 54

4.5.13, Line 6: Delete "except that resonance shall not be done".

PAGE 56

4.5.16d, Line 3: Delete "testing".

PAGE 57

4.5.17: Delete "within 10 milliseconds" and substitute "within 20 milliseconds".

4.5.19, Subparagraph a.: Delete “unloaded” and substitute “loaded”.

Subparagraph b.: Delete the first sentence in its entirety and substitute the following

“The same S&As shall be placed in a fuze assembly modified as follows:
(1) Use an Arming Mechanism Housing reworked for a slip-fit into the fuze body.
(2) The fuze assembly will not include a firing pin, firing pin spring nor locking balls.”

Subparagraph c.: Delete "DRDAR-QAR-M and DRDAR-QAR-I" and substitute "AMSTA-AR-QAC-S".

PAGE 58

6.3, Last Line: Delete "DRDAR-QAR-I" and substitute "AMSTA-AR-QAC-S".

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PAGE 59

6.6, Line 4: Delete "DRDAR-QAR-M" and substitute "AMSTA-AR-QAC-S".

PAGE 60

Appendix A: Delete in its entirety and substitute enclosed Figure 1.

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One 15 minute sinusoidal sweep (5-500-5 Hz) shall be performed for each of the three axes. No dwell time shall be spent on any one frequency. A search for resonance shall not be done.

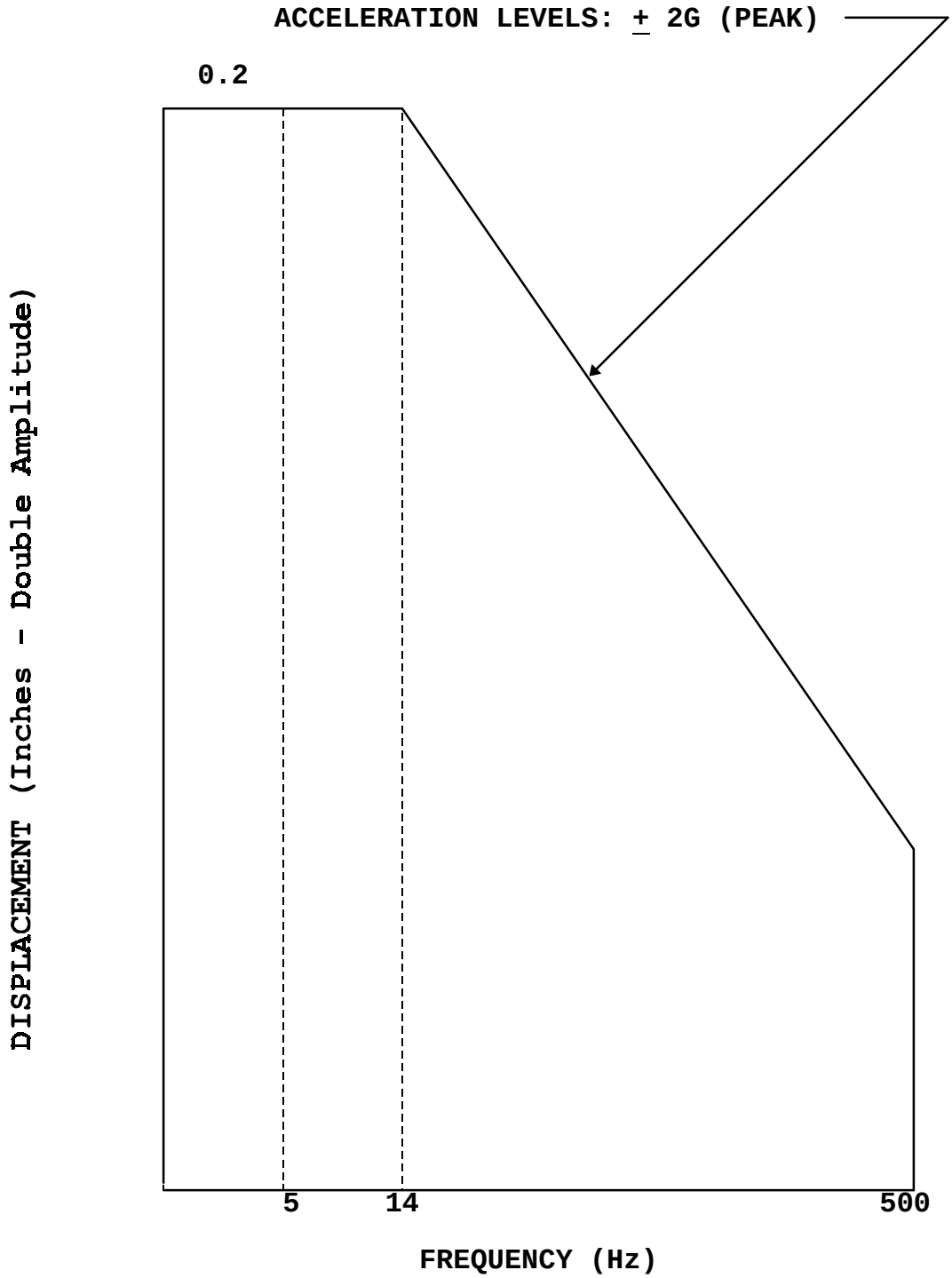

FIGURE 1. Helicopter vibration.

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